

Assignment 21

Q1. Create a Java application to manage employee records using CRUD operations

Java Code:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class Q1_ConnectionStatus {

    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/SIT_CSE";
        String user = "root";
        String password = "root";

        try {
            // Load Driver
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Establish Connection
            Connection con = DriverManager.getConnection(url, user, password);

            // Display Connection Status
            if (con != null && !con.isClosed()) {
                System.out.println("Database Connection Status: ACTIVE /
CONNECTED");
            } else {
                System.out.println("Database Connection Status:
DISCONNECTED");
            }

            // Close Resource
            con.close();

        } catch (ClassNotFoundException e) {
            System.out.println("Driver Class Not Found: " + e.getMessage());
        }
    }
}
```

```
    } catch (SQLException e) {  
        System.out.println("Connection Failed: " + e.getMessage());  
    }  
}
```

SQL Code:

```
-- Step 1: Create Database  
CREATE DATABASE IF NOT EXISTS SIT_CSE;  
  
-- Step 2: Use Database  
USE SIT_CSE;
```

Output:

```
Database Connection Status: ACTIVE / CONNECTED
```

Q2. Create a Java application to manage student records using CRUD operations. Perform Create, Read, Update, and Delete operations on student details such as roll number, name, course, and marks using a database

Java Code:

```
import java.sql.Connection;  
import java.sql.DriverManager;  
import java.sql.SQLException;  
  
public class Q2_StudentDatabaseConnect {  
  
    public static void main(String[] args) {  
        String url = "jdbc:mysql://localhost:3306/SIT_CSE";  
        String user = "root";  
        String password = "root";  
  
        try {  
            // Load Driver
```

```

Class.forName("com.mysql.cj.jdbc.Driver");

// Establish Connection
Connection con = DriverManager.getConnection(url, user, password);

// Display Specific Output Message
if (con != null && !con.isClosed()) {
    System.out.println("Connection Status: SUCCESS");
    System.out.println("Student database is connected successfully!");
}

// Close Resource
con.close();

} catch (ClassNotFoundException e) {
    System.out.println("JDBC Driver missing: " + e.getMessage());
} catch (SQLException e) {
    System.out.println("Database Connection Error: " + e.getMessage());
}
}
}

```

SQL Code:

```

-- Step 1: Select Database
USE SIT_CSE;

```

```

-- Step 2: Create Student Table
CREATE TABLE IF NOT EXISTS student (
    RollNo INT PRIMARY KEY,
    Name VARCHAR(100),
    Course VARCHAR(50),
    Marks DOUBLE
);

```

```

-- Step 3: Insert Initial Data
INSERT INTO student (RollNo, Name, Course, Marks) VALUES
(1, 'Rahul', 'CSE', 78.5),
(2, 'Aman', 'CSE', 85.0);

```

Output:

```
Connection Status: SUCCESS  
Student database is connected successfully!
```